

Kinematics in One and Two Dimensions

Describing motion precisely: where something is, how fast it moves, and how that changes, in one and two dimensions.

By **Pratik Bhanuse**, PhD Physics candidate, Western University ·
pratikbhanuse.github.io

Kinematics is the description of motion, without asking what causes it. That question, the *dynamics*, comes in Chapter 3. Here we build the vocabulary of position, displacement, velocity and acceleration. We learn to read motion from graphs, solve problems with constant acceleration, and extend everything to two dimensions with projectiles and relative motion. Throughout this chapter we treat moving objects as **particles**: points with no size.

YOU WILL BE ABLE TO

- Distinguish distance from displacement, and speed from velocity.
- Define and calculate average and instantaneous velocity and acceleration.
- Read and sketch $x-t$, $v-t$ and $a-t$ graphs, using slopes and areas.
- Derive and apply the constant-acceleration equations, including free fall.
- Handle acceleration that depends on time, velocity or position, using calculus.
- Analyse projectile motion: time of flight, maximum height, range and trajectory.
- Solve relative-velocity problems (rivers, rain, moving observers).

THINK FIRST

You drive to a town 60 km away at 30 km/h, then drive back at 60 km/h. What is your average speed for the round trip? (It is not 45 km/h.)

Going takes $60/30 = 2$ h and returning takes $60/60 = 1$ h. The total distance is 120 km in 3 h, so the average speed is **40 km/h**. Average speed is *total distance* $\div$ *total time*, not the average of the two speeds. You spend longer at the slower speed, so it counts for more.

§2.1

Position, displacement and distance

To describe motion along a straight line, choose an **origin** and a **positive direction**. Call the line the **x -axis**. The **position x** of a particle is its coordinate, and it can be negative.

The **displacement** over a time interval is the change in position:

$$\Delta x = x_{\text{final}} - x_{\text{initial}}.$$

Displacement is a vector. In one dimension its sign gives its direction.

The **distance** travelled is the total length of the path, a scalar that is never negative. Distance equals the magnitude of the displacement only if the particle never reverses direction.

For example, walk 5 m east and then 2 m west. The displacement is **+3 m** (3 m east), but the distance is 7 m.

§2.2

Velocity and speed

Average velocity and average speed

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}, \quad \text{average speed} = \frac{\text{total distance}}{\Delta t}.$$

Average velocity can be zero for a round trip, even though average speed cannot be. On an **x - t** graph, $\bar{v}$ is the slope of the **chord** joining the two points.

Instantaneous velocity

Shrink the interval to zero:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}.$$

On an **x - t** graph, **v** is the slope of the **tangent**. The **speed** is $|v|$, which is what a speedometer shows.

WORKED EXAMPLE 2.1

Average speed and average velocity

A cyclist rides 4.0 km east in 10 min, rests for 5 min, then rides 1.0 km west in 5 min. Find (a) the average speed and (b) the average velocity for the whole trip, in m/s.

Solution

Total time: $\Delta t = 10 + 5 + 5 = 20 \text{ min} = 1200 \text{ s}$.

(a) The total distance is $4.0 + 1.0 = 5.0 \text{ km} = 5000 \text{ m}$, so

$$\text{average speed} = \frac{5000 \text{ m}}{1200 \text{ s}} = 4.2 \text{ m/s}.$$

(b) Take east as positive. The displacement is $+4.0 - 1.0 = +3.0 \text{ km} = +3000 \text{ m}$, so

$$\bar{v} = \frac{+3000 \text{ m}}{1200 \text{ s}} = +2.5 \text{ m/s} \quad (2.5 \text{ m/s east}).$$

Note. The rest period counts in the total time, even though no distance is covered during it.

§2.3

Acceleration

Acceleration is the rate of change of velocity:

$$\bar{a} = \frac{\Delta v}{\Delta t}, \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}.$$

Its SI unit is m/s^2 . On a v - t graph, a is the slope of the tangent.

COMMON MISTAKE

“Negative acceleration” does not always mean “slowing down”. A particle **speeds up** when v and a have the **same sign**, and **slows down** when they have **opposite signs**. A car moving in the $-x$ direction with $a < 0$ is speeding up.

§2.4

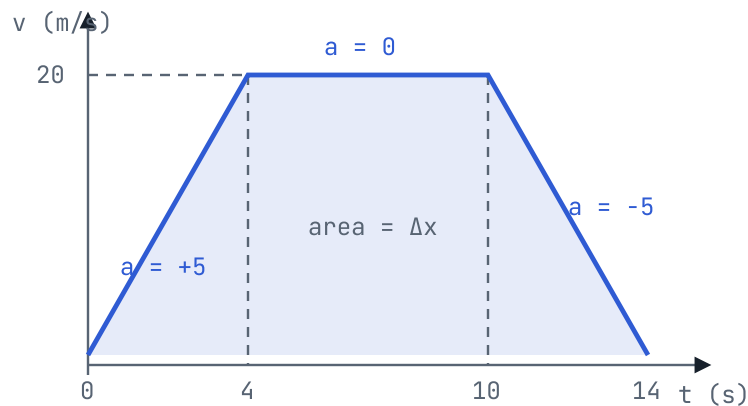
Motion graphs

Graphs are often the fastest way to understand a motion problem. Two facts carry almost everything.

KEY IDEA

- **Slopes:** the slope of $x-t$ is v , and the slope of $v-t$ is a .
- **Areas:** the area under $v-t$ is the displacement Δx , and the area under $a-t$ is the change in velocity Δv . Area below the time axis counts as negative.

These follow from calculus: $\Delta x = \int v dt$ and $\Delta v = \int a dt$.



A $v-t$ graph. The slope of each segment is the acceleration, and the shaded area is the displacement (Example 2.2).

WORKED EXAMPLE 2.2

Reading a velocity–time graph

A train's motion is shown in the graph above. Find (a) the acceleration in each stage, (b) the total distance travelled, and (c) the average velocity over the 14 s.

Solution

(a) Find the slope of each straight segment:

$$a_1 = \frac{20 - 0}{4 - 0} = 5 \text{ m/s}^2, \quad a_2 = 0, \quad a_3 = \frac{0 - 20}{14 - 10} = -5 \text{ m/s}^2.$$

(b) The distance is the area under the graph, a trapezium made of two triangles and a rectangle:

$$\Delta x = \frac{1}{2}(4)(20) + (10 - 4)(20) + \frac{1}{2}(14 - 10)(20) = 40 + 120 + 40 = 200 \text{ m.}$$

Alternatively, use the trapezium formula:

$$\frac{1}{2}(\text{sum of parallel sides}) \times \text{height} = \frac{1}{2}(14 + 6)(20) = 200 \text{ m. } \checkmark$$

(c) Average velocity:

$$\bar{v} = \frac{200 \text{ m}}{14 \text{ s}} = 14.3 \text{ m/s.}$$

§2.5

Motion with constant acceleration

Many important motions have constant acceleration: free fall near the Earth's surface, a car braking steadily, a charge in a uniform electric field.

Deriving the equations

Let the acceleration a be constant, and at $t = 0$ let the position be x_0 and the velocity v_0 (often written u).

Step 1. Integrate $dv/dt = a$:

$$v = v_0 + at. \tag{1}$$

Step 2. Integrate $dx/dt = v_0 + at$:

$$x - x_0 = v_0 t + \frac{1}{2}at^2. \tag{2}$$

Step 3. Eliminate t between (1) and (2). Substitute $t = (v - v_0)/a$ into (2) and simplify:

$$v^2 = v_0^2 + 2a(x - x_0). \tag{3}$$

Step 4. Because v is linear in t , the average velocity is the mean of the initial and final velocities:

$$x - x_0 = \frac{1}{2}(v_0 + v)t. \tag{4}$$

The same results follow from the v - t graph, which is a straight line. The area under it is a trapezium of parallel sides v_0 and v and width t , giving (4) directly.

KEY IDEA

For constant acceleration, with $s = x - x_0$ and $u = v_0$:

$$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u + v)t.$$

Each equation leaves out one of the five variables (s, u, v, a, t). Pick the equation that omits the quantity you neither know nor want.

A useful extra result is the **displacement in the n th second**, the distance covered between $t = n - 1$ and $t = n$:

$$s_n = u + \frac{a}{2}(2n - 1).$$

WORKED EXAMPLE 2.3

Stopping distance

A driver travelling at 72 km/h sees an obstacle. Her reaction time is 0.70 s, and then the brakes give a constant deceleration of 5.0 m/s^2 . Find the total stopping distance.

Solution

Identify. There are two stages: constant velocity during the reaction time, then constant deceleration.

Convert the speed: $u = 72 \text{ km/h} = 20 \text{ m/s}$.

Stage 1 (reaction). The car moves at constant speed:

$$s_1 = ut_r = (20 \text{ m/s})(0.70 \text{ s}) = 14 \text{ m}.$$

Stage 2 (braking). We know $u = 20 \text{ m/s}$, $v = 0$ and $a = -5.0 \text{ m/s}^2$, and we want s . Time is not needed, so use $v^2 = u^2 + 2as$:

$$0 = 20^2 + 2(-5.0)s_2 \Rightarrow s_2 = \frac{400}{10} = 40 \text{ m}.$$

Total: $s = 14 + 40 = 54 \text{ m}$.

Evaluate. The braking distance is proportional to u^2 . Doubling the speed to 144 km/h would quadruple the braking distance to 160 m, while the reaction distance only doubles

to 28 m.

IN PRACTICE

Crash physics and the design of cars. In a collision, injury depends on how large the occupant's *deceleration* is, and $v^2 = u^2 + 2as$ shows that stopping from a given speed over a **longer distance** means a **smaller** deceleration: $a = u^2/2s$. This is the whole idea behind crumple zones, seat belts that stretch slightly, and airbags. They turn a stop over a few centimetres (hundreds of g) into a stop over most of a metre (tens of g). Crash engineers quote decelerations in multiples of g . Sustained loads above roughly $50g$ to the chest, or brief peaks of a few hundred g to the head, are associated with serious injury. Problem P2.12 puts numbers on this.

Free fall

Near the Earth's surface, ignoring air resistance, every object falls with the same constant acceleration

$$g = 9.8 \text{ m/s}^2 \quad (\text{directed downwards}),$$

whatever its mass. With the y -axis pointing up, $a = -g$.

Useful results for a ball thrown straight up with speed u :

- Time to reach the top: $t_{\text{up}} = u/g$.
- Maximum height: $H = u^2/(2g)$.
- Time to return to the launch point: $2u/g$. The ball comes back with the same speed u , moving down.
- At the top, $v = 0$ but $a = -g$. The ball is not “weightless” there.

WORKED EXAMPLE 2.4

A stone thrown up from a cliff

A stone is thrown vertically upward at 15 m/s from the edge of a cliff 50 m above the sea. Find (a) the maximum height above the sea, (b) the time to reach the sea, and (c) the velocity just before it hits the water.

Solution

Set up. Put the origin at the launch point, with y up. Then $u = +15$ m/s and $a = -9.8$ m/s², and the sea is at $y = -50$ m.

(a) At the top $v = 0$:

$$0 = 15^2 - 2(9.8)y_{\max} \Rightarrow y_{\max} = \frac{225}{19.6} = 11.5 \text{ m.}$$

The height above the sea is $50 + 11.5 = 61.5$ m.

(b) Use $y = ut + \frac{1}{2}at^2$ with $y = -50$ m:

$$-50 = 15t - 4.9t^2 \Rightarrow 4.9t^2 - 15t - 50 = 0.$$

$$t = \frac{15 \pm \sqrt{15^2 + 4(4.9)(50)}}{2(4.9)} = \frac{15 \pm \sqrt{1205}}{9.8} = \frac{15 \pm 34.71}{9.8}.$$

Take the positive root: $t = 5.07$ s. The negative root (-2.01 s) is when a stone thrown upward from sea level *would* have passed the cliff top, which is not physical here.

(c) $v = u + at = 15 - 9.8(5.07) = -34.7$ m/s, that is, **34.7 m/s downward**. Check with $v^2 = u^2 + 2as = 225 + 2(-9.8)(-50) = 1205$, so $|v| = 34.7$ m/s. ✓

Evaluate. There is no need to split the motion into “up” and “down” parts. One set of equations with consistent signs handles the whole flight.

§2.6

When the acceleration is not constant

If a varies, the constant-acceleration formulas **do not apply**. Go back to the definitions and use calculus.

GIVEN	METHOD
$x(t)$	Differentiate: $v = dx/dt$, $a = dv/dt$.
$a(t)$	Integrate: $v = v_0 + \int_0^t a \, dt$, then $x = x_0 + \int_0^t v \, dt$.

GIVEN	METHOD
$a(v)$	Separate: $\frac{dv}{a(v)} = dt$ for $v(t)$, or $\frac{v dv}{a(v)} = dx$ for $v(x)$.
$a(x)$	Use $a = v \frac{dv}{dx}$: $\int v dv = \int a(x) dx$.

WORKED EXAMPLE 2.5

A boat slowing in water

A motorboat moving at $v_0 = 10$ m/s cuts its engine. Water resistance then gives it an acceleration $a = -kv$ with $k = 0.50 \text{ s}^{-1}$. Find (a) $v(t)$, (b) $x(t)$, and (c) the total distance the boat travels before stopping.

Solution

(a) Separate the variables in $dv/dt = -kv$:

$$\int_{v_0}^v \frac{dv}{v} = -k \int_0^t dt \Rightarrow \ln \frac{v}{v_0} = -kt \Rightarrow v = v_0 e^{-kt} = 10 e^{-0.5t} \text{ m/s.}$$

(b) Integrate $v(t)$ from $x = 0$ at $t = 0$:

$$x = \int_0^t v_0 e^{-kt} dt = \frac{v_0}{k} (1 - e^{-kt}) = 20 (1 - e^{-0.5t}) \text{ m.}$$

(c) As $t \rightarrow \infty$, $x \rightarrow v_0/k = 10/0.50 = 20$ m.

Shortcut. Use $a = v dv/dx$ directly: $v dv/dx = -kv$ gives $dv/dx = -k$, so v falls *linearly* with distance, $v = v_0 - kx$. The boat stops when $x = v_0/k = 20$ m. ✓

Evaluate. Strictly the boat never stops in finite time, since v decays exponentially, yet it covers only a finite distance. After $t = 6$ s, $v = 10e^{-3} = 0.5$ m/s and the boat has covered 95% of the 20 m.

§2.7

Motion in two dimensions

In two or three dimensions, position, velocity and acceleration become vectors:

$$\vec{r} = x\hat{i} + y\hat{j}, \quad \vec{v} = \frac{d\vec{r}}{dt} = v_x\hat{i} + v_y\hat{j}, \quad \vec{a} = \frac{d\vec{v}}{dt} = a_x\hat{i} + a_y\hat{j}.$$

The velocity is always **tangent to the path**. The acceleration need not be: it points towards the inside of a curving path.

KEY IDEA

Independence of perpendicular motions. The x and y motions are independent and linked only by the shared time t . If $\vec{a}$ is constant, each component separately obeys the one-dimensional constant-acceleration equations.

§2.8

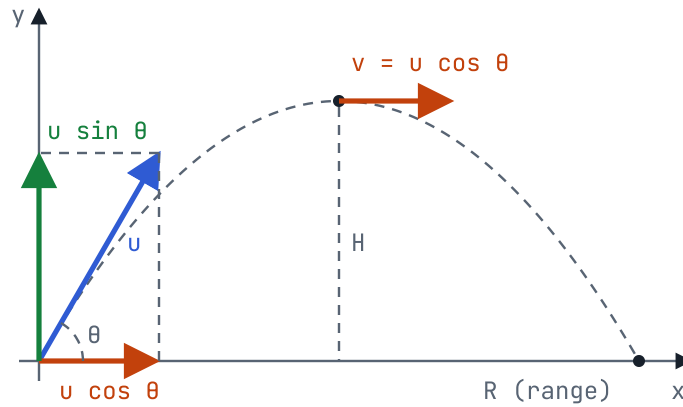
Projectile motion

A **projectile** is any object moving freely under gravity alone, with air resistance neglected. Take x horizontal and y vertically up, and launch from the origin with speed u at angle θ above the horizontal:

$$u_x = u \cos \theta, \quad u_y = u \sin \theta; \quad a_x = 0, \quad a_y = -g.$$

The equations of motion are:

Horizontal (constant velocity):	$x = u \cos \theta t,$	$v_x = u \cos \theta,$
Vertical (constant acceleration):	$y = u \sin \theta t - \frac{1}{2}gt^2,$	$v_y = u \sin \theta - gt.$



A projectile follows a parabola. The horizontal velocity stays constant at $u \cos \theta$. At the top the vertical velocity is zero.

Key results (launch and landing at the same height)

Time of flight. Set $y = 0$: $t(u \sin \theta - \frac{1}{2}gt) = 0$, so

$$T = \frac{2u \sin \theta}{g}.$$

Maximum height. At the top $v_y = 0$, so $0 = u^2 \sin^2 \theta - 2gH$, giving

$$H = \frac{u^2 \sin^2 \theta}{2g}.$$

Range. $R = u \cos \theta \cdot T = \frac{2u^2 \sin \theta \cos \theta}{g}$, so

$$R = \frac{u^2 \sin 2\theta}{g}.$$

The range is greatest at $\theta = 45^\circ$, where $R_{\max} = u^2/g$. Complementary angles (θ and $90^\circ - \theta$) give the same range, because $\sin 2\theta = \sin(180^\circ - 2\theta)$.

Trajectory. Eliminate $t = x/(u \cos \theta)$ from the y equation:

$$y = x \tan \theta - \frac{g x^2}{2u^2 \cos^2 \theta}.$$

This is a parabola, $y = bx - cx^2$, which proves that projectiles move on parabolic paths.

COMMON MISTAKE

The formulas for T , H and R assume that launch and landing are at the **same height**. For a launch from a cliff or onto a slope, go back to the component equations and solve them directly.

WORKED EXAMPLE 2.6**A football kick**

A football is kicked at 25 m/s at 37° above the horizontal. Take $\sin 37^\circ = 0.60$ and $\cos 37^\circ = 0.80$. Find (a) the time of flight, (b) the maximum height, (c) the range, and (d) the velocity of the ball 1.0 s after the kick.

Solution

Components: $u_x = 25(0.80) = 20$ m/s and $u_y = 25(0.60) = 15$ m/s.

(a) Time of flight:

$$T = \frac{2u_y}{g} = \frac{2(15)}{9.8} = 3.06 \text{ s.}$$

(b) Maximum height:

$$H = \frac{u_y^2}{2g} = \frac{225}{19.6} = 11.5 \text{ m.}$$

(c) Range:

$$R = u_x T = 20(3.06) = 61.2 \text{ m.}$$

Check: $u^2 \sin 2\theta / g = 625(2)(0.60)(0.80)/9.8 = 61.2 \text{ m. } \checkmark$

(d) At $t = 1.0$ s: $v_x = 20$ m/s and $v_y = 15 - 9.8(1.0) = 5.2$ m/s. So

$$v = \sqrt{20^2 + 5.2^2} = 20.7 \text{ m/s,} \quad \tan^{-1} \frac{5.2}{20} = 14.6^\circ \text{ above the horizontal.}$$

The ball is still rising, since $v_y > 0$.

WORKED EXAMPLE 2.7

Dropping supplies from a plane

A plane flying horizontally at 50 m/s, 80 m above flat ground, releases a supply package. Find (a) how long the package takes to land, (b) how far ahead of the release point it lands, and (c) its velocity on impact.

Solution

Identify. At release, the package has the plane's velocity: $u_x = 50$ m/s and $u_y = 0$.

(a) Vertical motion. Take y down, so $80 = \frac{1}{2}(9.8)t^2$ and

$$t = \sqrt{\frac{2(80)}{9.8}} = 4.04 \text{ s.}$$

(b) Horizontal motion: $x = 50 \times 4.04 = 202$ m.

(c) $v_x = 50$ m/s and $v_y = gt = 9.8 \times 4.04 = 39.6$ m/s downward. So

$$v = \sqrt{50^2 + 39.6^2} = 63.8 \text{ m/s,} \quad \tan^{-1} \frac{39.6}{50} = 38.4^\circ \text{ below the horizontal.}$$

Evaluate. Seen from the plane, the package falls straight down and stays directly below the plane, because both share the same horizontal velocity. This is relative motion, the topic of the next section.

IN PRACTICE

Real projectiles feel air resistance. The parabola is exact only in a vacuum. Air drag grows roughly as v^2 for everyday objects, so it shortens the range and makes the descent steeper than the ascent. A well-hit golf ball or baseball lands far short of its vacuum range, and the best launch angle falls below 45° . Engineers who design fire-hose nozzles, irrigation sprinklers, ski jumps and ballistic trajectories start from the ideal parabola and then add drag numerically. The parabola remains an excellent model for slow, dense objects over short distances: a thrown ball, a jet of water from a fountain, or a long jumper's centre of mass.

§2.9

Relative motion

Velocity is always measured **relative to a frame of reference**. If $\vec{v}_{AB}$ means “the velocity of A relative to B”, then velocities combine like this:

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}.$$

Read the subscripts as a chain: “A relative to C equals A relative to B plus B relative to C”. Two useful consequences:

- $\vec{v}_{AB} = -\vec{v}_{BA}$.
- $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$, where both velocities on the right are measured relative to the ground.

WORKED EXAMPLE 2.8

Crossing a river

A river 100 m wide flows at 3.0 m/s. A boat can move at 5.0 m/s relative to the water. (a) If the boat heads straight across, how long does the crossing take, and how far downstream does it land? (b) In what direction should it head to land directly opposite its starting point, and how long does that crossing take?

Solution

Set up. Take x downstream and y across the river. Write $\vec{v}_{bg} = \vec{v}_{bw} + \vec{v}_{wg}$ (boat-ground = boat-water + water-ground).

(a) Heading straight across: $\vec{v}_{bw} = 5.0 \hat{j}$ and $\vec{v}_{wg} = 3.0 \hat{i}$. Only the y component carries the boat across:

$$t = \frac{100 \text{ m}}{5.0 \text{ m/s}} = 20 \text{ s}, \quad \text{drift} = 3.0 \times 20 = 60 \text{ m downstream.}$$

(b) To land directly opposite, the boat’s velocity relative to the ground must have **no** x component. Head upstream at angle α from the straight-across direction, so that the upstream component of the boat’s water velocity cancels the current:

$$5.0 \sin \alpha = 3.0 \Rightarrow \sin \alpha = 0.60 \Rightarrow \alpha = 36.9^\circ.$$

The speed across is then $5.0 \cos \alpha = 4.0$ m/s, so

$$t = \frac{100}{4.0} = 25 \text{ s.}$$

Evaluate. Heading straight across is always the *fastest* crossing. Aiming upstream gives the *shortest path* but takes longer. If the current were faster than the boat, no heading could cancel the drift. See Problem P2.16.

WORKED EXAMPLE 2.9

Rain and an umbrella

Rain falls vertically at 8.0 m/s. A woman walks east at 6.0 m/s. At what angle should she tilt her umbrella, and how fast does the rain appear to fall to her?

Solution

The velocity of the rain relative to the woman is

$$\vec{v}_{rw} = \vec{v}_r - \vec{v}_w = (-8.0\hat{j}) - (6.0\hat{i}) = -6.0\hat{i} - 8.0\hat{j} \text{ m/s.}$$

To her, the rain moves downward and towards the west, so it appears to come from ahead of her and above. Its apparent speed is

$$|\vec{v}_{rw}| = \sqrt{6.0^2 + 8.0^2} = 10 \text{ m/s,}$$

at an angle $\tan^{-1}(6.0/8.0) = 36.9^\circ$ from the vertical. She should tilt her umbrella **forward (east)** by 36.9° from the vertical.

IN PRACTICE

Navigation is relative velocity. A pilot steers by the aircraft's velocity relative to the air, but arrives according to its velocity relative to the ground: $\vec{v}_{\text{ground}} = \vec{v}_{\text{air}} + \vec{v}_{\text{wind}}$. Flight planning software solves exactly the “river crossing” triangle of Example 2.8 to choose a heading that cancels a crosswind. The same vector bookkeeping is used for ships in ocean currents, drones in wind, and blood cells carried by flowing plasma past a Doppler ultrasound probe.

CHAPTER SUMMARY

- Displacement Δx is a vector. Distance is the path length.
- $v = dx/dt$ (slope of $x-t$) and $a = dv/dt$ (slope of $v-t$). The area under $v-t$ is Δx , and the area under $a-t$ is Δv .
- An object speeds up when v and a have the same sign and slows down when they have opposite signs.
- Constant a : $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s = \frac{1}{2}(u + v)t$. Free fall has $a = -g$ (with up positive).
- Variable a : use calculus, including $a = v dv/dx$.
- Projectiles: the horizontal velocity is constant and the vertical acceleration is $-g$.
 $T = 2u \sin \theta / g$, $H = u^2 \sin^2 \theta / 2g$, $R = u^2 \sin 2\theta / g$, and the path is a parabola.
- Relative velocity: $\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$ and $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P2.1

Can a body have zero velocity and yet a non-zero acceleration? Give an example.

P2.2

(a) Can a body move with constant speed while its velocity changes? (b) Can it move with constant velocity while its speed changes? Explain each.

P2.3

A ball is thrown vertically upward and caught at the same height. Taking upward as positive, sketch the $v-t$ and $a-t$ graphs for the whole flight. What is the slope of the $v-t$ graph?

P2.4

At the highest point of a projectile's path (launch angle between 0° and 90°), is the velocity zero? Is the acceleration zero? What are their directions?

LEVEL B – STANDARD PROBLEMS**P2.5**

A car starts from rest and accelerates uniformly to 30 m/s in 10 s. It then moves at constant speed for 20 s, and finally decelerates uniformly to rest in 5.0 s. Find the total distance travelled and the average speed.

P2.6

A stone dropped from the top of a tower falls 25 m in the last second before hitting the ground. Find the height of the tower.

P2.7

A particle starts at $x = 0$ with velocity 2.0 m/s and has acceleration $a = 6t$ (m/s², with t in s). Find its velocity and position at $t = 2.0$ s.

P2.8

A speeding car passes a parked police car at a constant 30 m/s. At that instant the police car starts from rest with a constant acceleration of **3.0 m/s²**. How long does it take to catch the speeder, how far does it travel, and how fast is it going at that moment?

P2.9

A ball is thrown at 20 m/s at 30° above the horizontal from the roof of a building 25 m tall. Find the time it takes to reach the ground and how far from the base of the building it lands.

P2.10

A projectile launched at 30 m/s lands 60 m away, at the same height it was launched from. Find the two possible launch angles and the maximum height reached in each case.

P2.11

Rain is falling vertically. A man walking at 4.0 km/h finds that the rain meets him at 30° to the vertical. Find the speed of the rain relative to the ground, and relative to the man.

P2.12

A car travelling at 54 km/h hits a rigid wall. Its front crumple zone collapses by 0.60 m as the passenger compartment stops. Assume constant decelerations throughout. (a) Find the deceleration of the passenger compartment (in m/s^2 and in multiples of g) and the duration of the crash. (b) An unbelted passenger keeps moving at 54 km/h until hitting the dashboard, which stops them in 5.0 cm. Find their deceleration in g . (c) A belted passenger with an airbag moves forward a further 0.30 m relative to the car while the car itself crumples, so their total stopping distance relative to the ground is 0.90 m. Find their deceleration in g .

LEVEL C – CHALLENGE PROBLEMS**P2.13**

A projectile is launched with speed u at angle θ above the horizontal, up a plane inclined at angle α (with $\theta > \alpha$). Show that its range along the incline is

$$R = \frac{2u^2 \cos \theta \sin(\theta - \alpha)}{g \cos^2 \alpha},$$

and find the angle θ that maximises it.

P2.14

A body entering a viscous liquid with speed v_0 experiences an acceleration $a = -kv^2$. Find (a) v as a function of distance x , (b) v as a function of time, and (c) the distance at which its speed has halved.

P2.15

A ball thrown vertically upward from the ground passes a height h at times t_1 (going up) and t_2 (coming down). Show that $h = \frac{1}{2}gt_1t_2$ and that the launch speed is $u = \frac{1}{2}g(t_1 + t_2)$.

P2.16

A river 120 m wide flows at 5.0 m/s. A boat can move at only 3.0 m/s relative to the water. In which direction should it head to be carried the *least* distance downstream while crossing? Find that minimum drift and the time taken.

P2.17

A small rocket is launched vertically from rest. Its engine gives it a constant upward acceleration of **20 m/s²** for 10 s, then shuts off. Neglecting air resistance, find (a) the speed and height at burnout, (b) the maximum height reached, and (c) the total time from launch until the rocket hits the ground.

P2.18

Traffic engineering. A traffic engineer must set the yellow-light time at an intersection 20 m wide, on a road with a 50 km/h speed limit. Drivers have a reaction time of 1.0 s and brake at **4.0 m/s²**. Cars are 5.0 m long. (a) Find the minimum distance from the stop line at which a driver at the speed limit can still stop. (b) A driver who is just too close to stop must instead continue at constant speed and clear the far side of the intersection completely before the light turns red. What is the minimum yellow time that avoids a “dilemma zone”, where a driver can neither stop nor clear?

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

© 2026 Pratik Bhanuse. All rights reserved. · pratikbhanuse.github.io